

Introduction to Pure Mathematics

Problem Sheet 1

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Part A

1. (a) $1, 2, 3, 4$ is not a set however $\{1, 2, 3, 4\}$ is a set.
 (b) $\{1, 2, 4, 3, 3\}$ is a set.
 (c) $\{1, \{2, 3\}, 4\}$ is a set.
 (d) $\emptyset$ is a set.
 (e) $\{\text{blue}, \text{house}\}$ is a set.
 (f) $\{\text{horse}, \text{cat}, \emptyset\}$ is a set.
2. (a) " $3 = 2$ " is a statement.
 (b) " $3 + 2$ " is not a statement.
 (c) " $3 \geq 2$ " is a statement.
 (d) " 3 and 2 " is not a statement.
 (e) " $\{4\} \subseteq \mathbb{Z}$ and $4 = 2$ " is a statement.
 (f) " $4 \in \mathbb{Z}$ or $4 + 2$ " is not a statement.
 (g) " $\mathbb{Z} \cap \mathbb{Q}$ " is not a statement.
 (h) " $\mathbb{Z} \subseteq \mathbb{Q}$ " is a statement.
 (i) " $|5|$ " is not a statement.
3. Let $A = \{\text{cat}, \{1, \text{cat}\}, \alpha, \{\alpha\}\}$
 - (a) $\text{cat} \in A$ is true.
 - (b) $\{\text{cat}\} \notin A$ is true.
 - (c) $\{1, \text{cat}\} \subseteq A$ is false.
 - (d) $\{\text{cat}, \alpha\} \subseteq A$ is true.
 - (e) $\emptyset \subsetneq A$ is true.
 - (f) $A = \{\alpha, \text{cat}, \{1, \text{cat}\}, \{\alpha\}\}$ is true.
 - (g) $A = \{1, \{1, \text{cat}, \}, \alpha\}$ is false.
4. Let $A = \{1, \sqrt{2}, \pi, \text{cat}\}$ and $B = \{0, \sqrt{2}, \text{dog}, \text{cat}\}$ then

$$A \cup B = \{1, \sqrt{2}, \pi, \text{cat}, 0, \text{dog}\}$$

$$A \cap B = \{\sqrt{2}, \text{cat}\}$$

$$A \setminus B = \{1, \pi\}$$

$$B \setminus A = \{0, \text{dog}\}$$
5. For the following 13 subsets of $\mathbb{R}$:
 $\emptyset$; $\mathbb{Z}$; $2\mathbb{Z}$; $A_1 = \mathbb{Z} \setminus \mathbb{R}$; $A_2 = \{x \in \mathbb{R} : 1 \leq x \leq 3\}$; $A_3 = \{1, 2, 3\}$; $A_4 = \mathbb{Z}_- \cap \{0\} \cap \mathbb{Z}_+$;
 $A_5 = \mathbb{Z}_- \cup \{0\} \cup \mathbb{Z}_+$; $A_6 = \{x \in \mathbb{Z} : x \geq 1 \text{ or } x \leq 3\}$; $A_7 = \{x \in \mathbb{Z} : x \geq 1 \text{ and } x \leq 3\}$;
 $A_8 = \{x \in \mathbb{Z} : \forall k \in \mathbb{Z}, x = 2k\}$; $A_9 = \{x \in \mathbb{Z} : \exists k \in \mathbb{Z} \text{ such that } x = 2k\}$; $A_{10} = \{x \in \mathbb{Z} : \exists k \in \mathbb{R} \text{ such that } x = 2k\}$

We have the following equal subsets

- $\emptyset = A_1 = A_4 = A_8$
- $\mathbb{Z} = A_5 = A_6 = A_{10}$
- $2\mathbb{Z} = A_9$
- A_2 is not equal to any other subset
- $A_3 = A_7$

We also have the following chains of subsets:

- $\emptyset \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}$
- $\emptyset \subsetneq A_3 \subsetneq \mathbb{Z}$
- $\emptyset \subsetneq A_3 \subsetneq A_2$

6. We know that $\mathbb{Z}_+ \subseteq \mathbb{Z}_{\geq 0} \subseteq \mathbb{Z}$. Since $0 \in \mathbb{Z}_{\geq 0}$ and $0 \notin \mathbb{Z}_+$ it must be true that $\mathbb{Z}_{\geq 0} \neq \mathbb{Z}_+$ and similarly $-1 \in \mathbb{Z}$ and $-1 \notin \mathbb{Z}_{\geq 0}$ we get that $\mathbb{Z} \neq \mathbb{Z}_{\geq 0}$. Therefore $\mathbb{Z}_+ \subsetneq \mathbb{Z}_{\geq 0} \subsetneq \mathbb{Z}$.
7. $A, B \subseteq \mathbb{Z}$
 - (a) $A \cap B = \emptyset$ and $A \cup B \neq \mathbb{Z}$ is true for $A = \emptyset$ and $B = \emptyset$
 - (b) $A \cap B \neq \emptyset$ and $A \cup B = \mathbb{Z}$ is true for $A = \mathbb{Z}$ and $B = \mathbb{Z}$
 - (c) $A \cap B = \emptyset$ and $A \cup B = \mathbb{Z}$ is true for $A = \emptyset$ and $B = \mathbb{Z}$
8. Let $\{A_i : i \in \mathbb{Z}_{\geq 0}\}$ be a collection of subsets of $\mathbb{Z}$ defined by $A_i = \{x \in \mathbb{Z} : -i \leq x \leq i\}$.
 - (a) $\bigcup_{i \in \{1,2,3,4\}} A_i = \{-4, -3, \dots, 0, \dots, 3, 4\}$
 - (b) $\bigcap_{i \in \{1,2,3,4\}} A_i = \{-1, 0, 1\}$
 - (c) $\bigcup_{i \in \mathbb{Z}_{\geq 0}} A_i = \mathbb{Z}$
 - (d) $\bigcap_{i \in \mathbb{Z}_{\geq 0}} A_i = \{0\}$
9. TODO

Part B

1. Let $A = \{x \in \mathbb{R} : x < 5\}$ and $B = \{\pi, 2\sqrt{5}, 5\}$ be subsets of $\mathbb{R}$
 - (a) $A \cap \mathbb{Z} = \{x \in \mathbb{Z} : x \leq 4\}$; $A \cap B = \{\pi, 2\sqrt{5}\}$; $A \cup B = \{x \in \mathbb{R} : x \leq 5\}$; $B \setminus A = \{5\}$; $A^c = \{x \in \mathbb{R} : x \geq 5\}$
 - (b) TODO
2.
 - (a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $xy = 1$ is a false statement.
 - (b) $\forall x, y \in \mathbb{R}$ we have $xy = 1$ is a false statement.
 - (c) $\exists x \in \mathbb{R}$ such that $\exists y \in \mathbb{R}$ such that $xy = 1$ is a true statement.
 - (d) $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}$ we have $xy = 1$ is a false statement.
3. TODO
4. TODO
5. TODO
6. TODO

7. Let $a, b \in \mathbb{R}$ with $a < b$. Show that $\frac{a+b}{2} < b$ and $a < \frac{a+b}{2}$. Arithmetic properties of $\mathbb{R}$ these are satisfied $\forall x, y, z \in \mathbb{R}$

$$(\mathbb{R}1) \ x \cdot 0 = 0, 1 \cdot x = x \text{ and } x + 0 = x$$

$$(\mathbb{R}2) \ -(-x) = x \text{ and } -0 = 0$$

$$(\mathbb{R}3) \ y + x = x + y \text{ and } x + (y + z) = (x + y) + z$$

$$(\mathbb{R}4) \ \text{if } x > y \text{ then } x + y > y + z; \text{ in addition, if } z > 0 \text{ then } xz > yz$$

$$(\mathbb{R}5) \ \text{if } x > y \text{ and } y > z \text{ then } x > z$$

$$(\mathbb{R}6) \ x^2 \geq 0$$

Proof: Starting with $a < b$ we can add b to both sides due to $\mathbb{R}4$ giving us $a + b < 2b$ then multiply both sides by $\frac{1}{2}$ since $\frac{1}{2} > 0$ giving us $\frac{a+b}{2} < b$. In order to prove the second statement we again start with $a < b$ then add a using $\mathbb{R}4$ giving $2a < a + b$ and again times through by $\frac{1}{2}$ giving $a < \frac{a+b}{2}$. Therefore $\forall a, b \in \mathbb{R}$ if $a < b$ then $\frac{a+b}{2} < b$ and $a < \frac{a+b}{2}$.

Part C

1. Find subsets A, B and C of $\mathbb{R}$ such that

$$(a) \ (A \cap B) \cup C \neq A \cap (B \cup C)$$

$$(b) \ (A \cap B) \cup C \neq A \cap (B \cup C)$$